Cyber Resilience in Solar-Rich Networked Microgrids: A Real Time End-to-End DER Outage Management Framework

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***Abstract*—The increasing integration of distributed energy resources (DERs), especially solar PV with distribution automation and inherent cyber-vulnerabilities, introduces new challenges in outage detection, diagnosis, and recovery due to integrated cyber-physical dependency. Traditional outage management systems (OMS) focus primarily on load-side faults and overlook DER-specific outages, particularly those caused by cyber-induced anomalies in the data or control planes. This work proposes a cyber-resilient DER Outage Management System (DEROMS) tailored for solar-rich, networked microgrids, considering the increasing integration of solar generation within distribution systems among DERs. The developed framework includes a TraceAlign-DPI module for real-time data integrity checking and hidden outage detection, a TraceAlign-RCA engine for multi-source root cause inference, and scenario-specific restoration optimization leveraging networked microgrid flexibility. A testbed built in Real-Time Digital Simulator (RTDS) validates the developed framework for physical faults and cyberattack scenarios. Results show improved accuracy in outage classification and enhanced system resilience through adaptive restoration under uncertainty.**

***Index Terms*—Cyber resilience, networked microgrids, outage management, real-time testbed, synchrophasor data.**

I. Introduction

A. Background and Motivation

THE distribution system is rapidly evolving with the rise of microgrids and renewable-based DERs. Once niche, microgrids are now a key resilience strategy for utilities like PG&E, SDG&E, and ComEd, offering islanding and local control capabilities [1]. The solar PV microgrid market alone is expected to top USD 25 billion by 2034, driven by electrification and decarbonization goals [2]. The U.S. DOE has recognized microgrids as vital for enhancing reliability, efficiency, and emissions reduction, targeting 98% outage reduction and 20% efficiency gains over conventional systems [3]. Meanwhile, utility-scale microgrids are projected to exceed USD 50 billion [4].

At the same time, utility-scale solar PV has become the leading source for new capacity additions in the U.S., adding 20.2 GW in 2022 alone, surpassing investments in other DERs such as wind and battery storage [5]. Falling costs and supportive policies are pushing distributed and grid-scale solar deeper into the distribution system [6], [7], [8], [9], [10]. A critical enabler of these trends is the proliferation of smart inverters embedded in solar PV and battery storage systems. They allow DERs to deliver voltage regulation, frequency support, and ride-through capabilities, turning them into active grid assets. These features enable Non-Wires Alternatives (NWA), helping defer costly grid upgrades while improving flexibility [11], [12], [13], [14], [15], [16]. Moreover, these renewable resources can provide critical self healing capabilities through potential formation of networked microgrids (NMG) and coordinated energy management [17].

As DER penetration grows, ensuring their secure and resilient operation becomes critical for grid stability, which has led to significant improvements in visibility and observability, driven by widespread sensor deployment and advances in communication protocols [18], [19], [20]. Real-time data streams from smart inverters, PMUs, advanced meters, and grid-edge monitoring devices now enable utilities to better monitor and control DER operations [21]. However, the very features that enhance DER operation - namely, remote monitoring and remote controllability simultaneously expand the cyber failures, including expansion of the cyberattack surface, exposing DER assets to a range of potential cyber-physical manipulations and failures [22], [23]. Traditionally, Outage Management Systems (OMS) under Advanced Distribution Management Systems (ADMS) were designed primarily to address load outages, as distribution systems historically operated as passive networks where load was the critical component. Yet, with the increasing integration of DERs, particularly utility-scale DERs, these resources have become equally critical. Unmonitored or unplanned DER outages can now trigger cascading failures, resulting in severe operational disruptions and significant economic losses

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[24]. The DER Outage Management System (DEROMS) can therefore be integrated within a Microgrid Management System (MGMS), which usually has full observability of internal DERs, or DER management system (DERMS) or ADMS. Reports and industry analyses consistently stress that while DERMS, MGMS or ADMS are maturing, DEROMS must also evolve to address these emerging cyber-physical risks [24], [25].

In existing literature, few works related to DER outages have primarily focused on detecting and managing outages originating from physical faults. There is a pressing need to transition toward cyber-informed DEROMS capable of real-time monitoring, outage detection, and outage classification that spans both physical and cyber domains. Such a system must not only identify and diagnose the root cause of an outage to be operational or cyber-induced, but also tailor appropriate, scenario-specific response strategies specially utilizing the potential of networked microgrids. Motivated by this gap, the current work proposes a real-time end-to-end outage management framework designed to leverage multi-domain DER data streams, detect and diagnose both physical and cyber induced outage scenarios, identify the root cause of disruptions, and enable adaptive, cyber-resilient response measures incorporating flexibilities of networked microgrids suited for real-time or near real-time application.

B. Literature Review and Research Gap

While traditional outage detection methods rely heavily on trustworthy measurements and sensor data, it is critical to first address data plane inconsistencies. Several existing approaches effectively implemented real-time integrity checks by identifying deviations from expected "normal" behavior during system operation based on PMU measurements, fault alarms, SCADA, and/or breaker status [26], [27]. However, a common vulnerability in these methods is the implicit assumption that the reported as well as historical data used for model training is inherently uncompromised. These event detection tools may provide false positives if the reported data is falsified and there is a lack of a distinct, preliminary validation stage to trust this baseline data before it is used to build or run the detection algorithms, which leaves the models susceptible to being trained on corrupted data. Furthermore, even methods that monitor high-level network parameters, such as aggregate power flow patterns, to detect database anomalies [28] often lack the granularity to address device-specific events. A data-driven cyber physical detection model uses PMU coherence features and a stack of Deep belief networks to flag corrupted measurements in real time [29], an isolation forest approach applies PCA for dimensionality reduction, followed by an unsupervised forest model to isolate anomalous patterns [30], and an OPF-forecast deviation method continuously compares live OPF outputs against a statistical forecast of expected power flow patterns to detect database tampering [28]. However, none of these works perform a separate, standalone assessment of its historical "natural" (attack-free) dataset before model training. The sanity check occurs only as part of the anomaly-detection or forecasting process itself. Furthermore, a key challenge remains unaddressed: the inability to confirm the genuine operational status of individual DERs as a whole. To overcome this, we need an integrated detection framework that combines broad network analysis with granular, device-level sensor measurements.

For outage root cause analysis, solar-rich systems present unique challenges because these resources are both generation assets and cyber-physical systems. Most existing work on solar DER outage root cause inference focuses on PV fault detection and classification [31], [32], [33], [34], [35], [36], [37]. Authors in [38] review islanding detection techniques, whereas [39] proposes a sensor failure detection mechanism. These studies, although valuable for identifying PV system and inverter faults, only comprise a subset of outage types and often exclude the systematic diagnosis of cyber-physical vulnerabilities under complex protection schemes [40], [41], [42]. Authors found two works [43], [44] that account for cyber-induced scenarios behind outage, but implement the strategy primarily on load outages. Though the work in [45] advances large scale cyber network simulation and detects critical cyber nodes, it has limited applicability towards cyber-aware restoration strategies or DER outage from a power system perspective.

Again, in evolving outage mitigation strategies, networked microgrids are becoming popular due to their resilient service restoration post-disturbance [46], [47]. These works optimize load recovery and network reconfiguration but largely ignore cyber-physical threats. For instance, they assume reliable DER operation and neglect compromised assets or data integrity loss during the restoration planning process.

To summarize, there is a lack of holistic, system-wide perspective that integrates different operational layers and a complete end-to-end DER outage management framework. Most data falsification detection techniques are not tailored for DER outage scenarios. Few works offer integrated cyber-aware outage RCA, and almost none integrate this insight into restoration decision-making in the presence of DERs, creating a dire need for a comprehensive framework that can enforce true system-wide coherence by aligning data traces from both cyber-physical domains. The key gaps are as follows,

- Lack of a comprehensive related work on possible outage scenarios in emerging cyber-physical systems from DER perspective.
- Outage management remains limited primarily focusing on physical faults rather than root-cause resilience across data/ control failure modes.
- Lack of system-wide, cross-domain trace alignment, limiting the detection of sophisticated masked or unobserved DER outages.
- No end-to-end framework that integrates detection, RCA, and restoration under both physical and cyber disruption scenarios for solar-rich DER systems.
- Restoration strategies in networked microgrids seldom incorporate cyber-awareness, leading to sub-optimal restoration or risk amplification.

C. Contributions

This work proposes a real-time DER outage management framework for solar-rich networked microgrids that is both *cyber-aware* and *restoration-oriented*. While many recent works apply ML methods to largely target individual functions, our proposed DEROMS emphasizes the interaction and hierarchy of detection, RCA, and restoration modules, with ML techniques embedded as support tools in a coordinated manner. The contributions are summarized as,

1) Analyzed DER outage scenarios considering cyber failure in data and control planes.
2) Proposed an end-to-end DER outage management framework, identifying vital tools, defining their specific functions and proposing precise hierarchy on how these tools will be triggered and coordinated.
3) Proposed TraceAlign-DPI, a DEROMS framework that ensures data plane integrity by aligning cross-domain traces while using forecast-aware estimation to compensate for DER visibility loss.
4) Proposed TraceAlign-RCA, a DEROMS framework to align cross-domain pre-outage traces to distinguish fault induced vs. cyber-induced outages.
5) Discussed restoration strategies informed with improved situational awareness on DERs based on previous outage detection and RCA layers.
6) Developed real-time testbed of networked microgrid with power electronics and protection interfaces as well as real-time data acquisition and monitoring.
7) Developed two case studies to demonstrate the workflow of the DEROMS on a cyber-compromised vs. normal scenario.

The proposed end-to-end DEROMS framework is classified into three functional layers namely - Outage Detection Layer, Outage Root Cause Analysis Layer and Outage Response and Recovery Layer. The detailed description of each of these layers is discussed in the following three sections with the complete architecture in Fig. 1.

II. Outage Detection Layer

A. Understanding Outage Scenarios Considering Cyber Failure in Data Plane

In an ideal scenario, reliable and synchronized streaming data from DER-end monitoring devices, such as PMUs, SCADA gateways, smart meters, breaker status, would allow precise real-time detection of the DER status. Throughout this work, the term **DER status** refers to either connected and supplying power or offline/facing outage. However, real-world cyber-physical systems are prone to cyber failures that degrade or distort system observability. These failures, located in the data plane, can arise from multiple sources such as sensor malfunctions or calibration drift, communication errors, including dropped or corrupted packets, complete or partial loss of data transmission or attack-induced falsification of measurement data streams. In such conditions, the outage detection layer must account for a range of possible visibility scenarios that may occur during DER monitoring:

1) *Visible True Outage:* A physical outage occurs at the DER, and the event is correctly reflected in the incoming data streams. This is the ideal case where outage detection operates as expected.
2) *Falsely Reported Outage:* A DER remains operational, but due to data plane failures (such as erroneous measurements, corrupted packets, or falsified data injection), an artificial loss of power is reported in the monitoring data, incorrectly indicating that a DER outage has occurred.
3) *Masked Outage:* A DER experiences a physical outage, but the event is not reported or captured in the data stream due to complete sensor inaccuracies, delayed measurements, or data masking attacks that suppress outage signatures.
4) *Unobserved Outage due to Loss of Data / Visibility:* The monitoring system loses all or most communication with the DER, resulting in total visibility loss. The DEROMS cannot ascertain the DER's current operational status under this condition.

Traditional DERMS and MGMS architectures primarily rely on observing direct data indicators to declare an outage, which limits their reliability in non-ideal monitoring conditions. In contrast, a cyber-resilient DER Outage Management System (DEROMS) must be designed to distinguish true physical outages from data plane anomalies, detect inconsistencies across multiple data sources, and operate under degraded or adversarial observability conditions.

B. Proposed Cyber-Resilient Outage Detection & Classification Architecture With TraceAlign-DPI

The outage detection mechanism in the proposed framework incorporates both conventional outage indicators and advanced data plane integrity verification to improve resilience against cyber-physical inconsistencies. The central module within the detection layer is the proposed **TraceAlign-DPI (Cross Domain Outage Trace Alignment for Data Plane Integrity)** tool, which operates both as a reactive and proactive data integrity evaluator across the operational states of the DER. Three other tools are incorporated in the framework named **Solar Forecasting tool**, **Forecast-Aware Data Estimation tool**, and **Cyber Anomaly Detection tool**. Hence a brief description of the purpose of these tools as well as how they interact with TraceAlign-DPI to enhance the DEROMS framework is added in this section.

1) Functional Objective of TraceAlign-DPI: TraceAlign-DPI is a lightweight, event-triggered or periodic data plane integrity checking module embedded inside MGMS, designed to cross-validate DER observability data streams with system level dynamics to ensure sensor-level data is consistent and coherent with the system before outage RCA or response actions are taken. TraceAlign-DPI aims to verify whether the operational status reported by a DER (either as connected and supplying power or as tripped/ offline) is consistent with the dynamic behavior of the surrounding system as observed via system-wide monitoring. The module utilizes the physics of power system balance: any real DER outage will alter the load-generation equilibrium, thus inducing measurable changes in network-wide variables commonly manifesting through shifts/ anomalies in SCADA-measured PQ flows across feeders, loads, and neighboring DERs or in frequency, ROCOF and voltage profile from

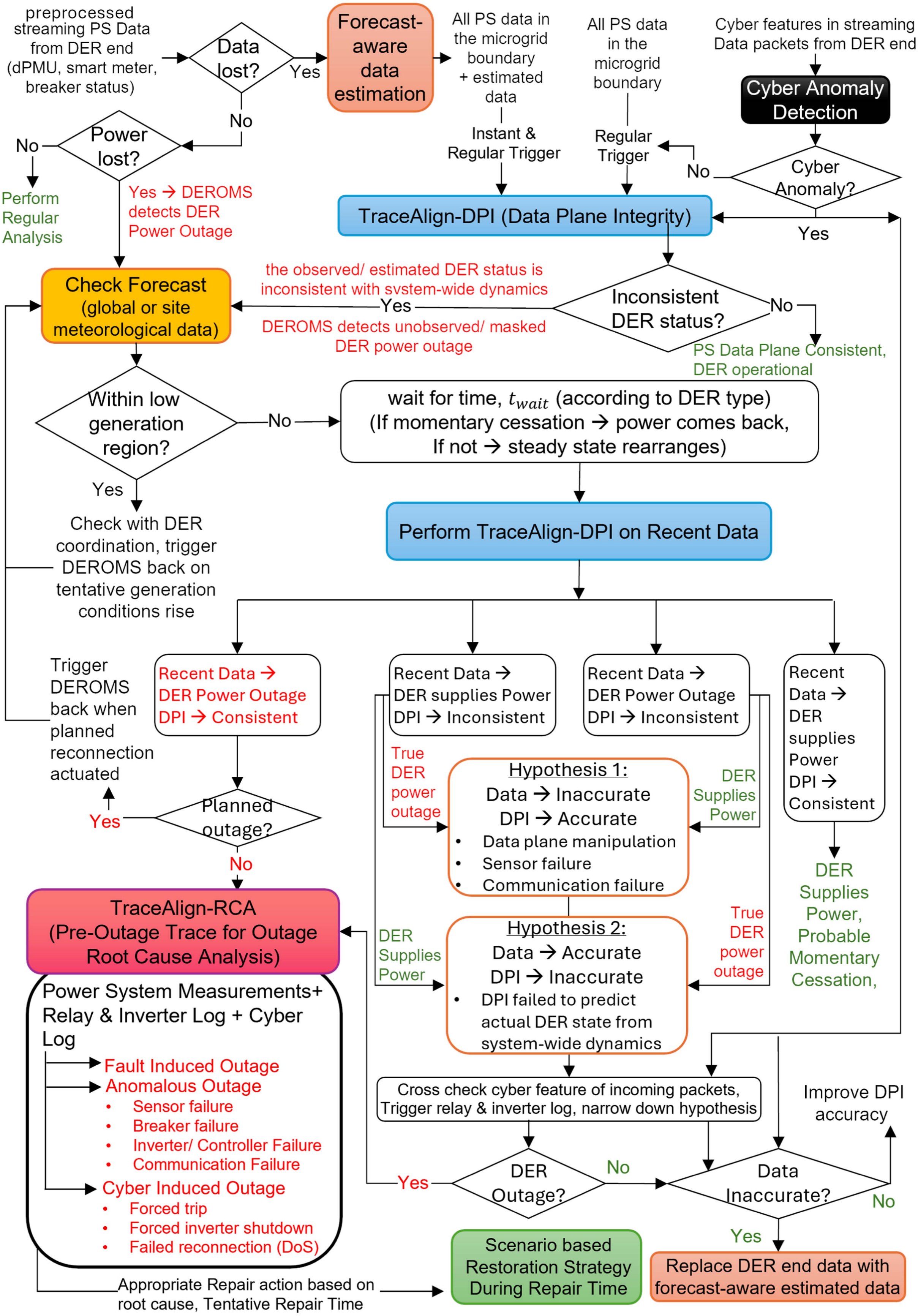

Fig. 1. Proposed architecture of end-to-end deroms framework.

PMU measurements. Three thresholds are defined in this case: T_p, primary anomaly threshold - above which a system anomaly/ shift indicates possible outage and raises a positive outcome of the tool, $T_{ls}, secondary\ lower\ threshold$ - below which system shift is considered too low to have incurred a DER outage and $T_{us},\ secondary\ upper\ threshold$ - above which means very strong post-outage effect detected. T_{us} and T_{ls} are needed when data-DPI contradicts. By aligning the reported DER status against system-wide dynamic signatures, TraceAlign-DPI establishes whether the DER-side data is trustworthy or possible data plane corruption exists.

2) Operational Deployment of TraceAlign-DPI: TraceAlign-DPI operates in the following modes within DEROMS:

a) Invocation Driven by DER Power Loss Detection (Reactive Mode 1): Upon detection of a DER power loss via streaming power system (PS) measurements, it is first checked whether the power loss is expected due to weather conditions (i.e. low irradiance region) by utilizing a **solar forecasting tool**. The Solar Forecasting tool utilizes site specific or area based meteorological data to nowcast a range of tentative solar generation. The tool leverages varying irradiance profiles to account for natural solar intermittency, ensuring that power loss from low irradiance is not misclassified as an outage. Also, to account for second-level variability, a tolerance band around the forecasted value should be considered where measured PV power within this band is attributed to irradiance changes rather than outages. If the generation forecast is not in the low generation region but power loss was reported, TraceAlign-DPI is triggered after a wait time t_{wait} to validate the authenticity of the detected outage event. An explanation of t_{wait} is added at the end of this subsection. The consistency check proceeds as follows:

- If the reported DER power loss is corroborated by system-wide shifts consistent with load-generation rebalancing, then *true DER outage* is predicted.
- If inconsistency is observed, that is, the DER reports power loss while the system shows no expected system wide response - this suggests possible data plane corruption or *falsely reported outage* (e.g., falsified zero-power reading, data masking). The proposed framework tries to detect a system shift with a lowered secondary anomaly threshold as well as evaluating the trustworthiness of the incoming packets through a **cyber anomaly detection tool** to confirm the actual DER status. If outage is predicted through the combined output of these tools, relay/ inverter log is invoked for the root cause of outage. This adds an additional confidence on DER outage detection and if data is found to be unreliable and inaccurate, DER side data is replaced by forecast aware estimated data for continued analysis.

b) Invocation Triggered by Cyber-Layer Anomaly Detection (Reactive Mode 2): TraceAlign-DPI is also triggered though a cyber anomaly detection tool upon the detection of cyber anomalies observed in the metadata or protocol layers of incoming streaming packets from dPMU and SCADA sources even if the PS data shows no power loss from the DER end. These packets traverse multiple network layers including Ethernet, IP, transport protocols (TCP/UDP), and application-layer protocols such as IEEE C37.118, DNP3, or Modbus. While their payloads contain physical system measurements, the surrounding protocol headers embed cyber-relevant metadata such as:

- IP/MAC address mismatches compared to expected
- Unexpected function codes or command flags
- Missing or altered CRC values
- Irregular sequence numbers or timestamp fields
- Abnormal response delays or jitter patterns

These diagnostics distinguish between: Non-critical anomalies (e.g., jitter, retransmission) which do not question payload trustworthiness, and Critical anomalies which directly question the credibility of DER measurements. Malicious events may leave metadata traces (e.g., abnormal source/ destination IP/ port, sequence number gaps, replayed timestamps, checksum errors) that the cyber anomaly tool can flag as critical anomalies. In contrast, communication loss or sensor failures which are also critical cyber anomalies may have metadata features (e.g., missing sequence counters, link-layer flags, quality indicators in PMU streams) which indicate degraded data integrity. Perhaps, considering practical limitations and exclusivity of the cyber configuration, these anomalies may not always manifest as explicit metadata violations or be clearly distinguishable as critical vs. non-critical. Therefore, our framework uses metadata anomalies to provide early suspicion of integrity threats, but final classification integrates both metadata diagnostics and system-level Data and DPI findings.

Thus, upon observing critical cyber-layer inconsistencies, the DEROMS invokes TraceAlign-DPI to validate the power system measurements themselves. When the payload containing the PS measurements shows DER power supply, a consistent DER status indicates coherent system-wide dynamics even though cyber anomaly exists. An inconsistency by TraceAlign-DPI would mean that, the system reflects a possible DER outage though the DER end data shows power supply- *masking an outage*, further reinforcing the suspicion that the data stream is corrupted.

c) Periodic Invocation (Proactive Mode): Independently of event triggers, TraceAlign-DPI is periodically called as a continuous data integrity monitor. In this mode:

- The system predicts expected DER output based on real-time network-wide measurements and compares it with reported DER status.
- If the DER reports being active, but system-wide data suggests power imbalance consistent with an outage, TraceAlign-DPI flags a *masked/hidden DER outage* when a very strong system wide shift is observed (system anomaly score (AS)$> T_{us}, secondary\ upper\ threshold$). Again, checking with cyber traces and invoking relay and inverter logs can confirm or discard the finding of TraceAlign-DPI.

This mechanism enables detection of silent outages where no direct trip signal is reported due to data falsification.

d) Invocation Under Data Loss Conditions (Visibility Loss Compensation): As per previous discussion, upon investigation through TraceAlign-DPI, cyber anomaly tool and further relay/inverter log, when DEROMS confirms that data does not reflect the system (i.e. data is falsified or lost), it flags the scenarios as data visibility loss. In these cases, DEROMS substitutes the missing data with forecast-based synthetic estimation through **Forecast-aware data estimation tool**. This tool assumes the DER continues to operate at its last-known status on forecasted capacity. Before the visibility returns, estimation followed by TraceAlign-DPI is periodically invoked to keep verifying:

- Whether the system-wide measurements remain consistent with the synthetic estimate.
- Whether an abrupt deviation suggests the DER has actually tripped while remaining unobservable to predict an *unobserved outage*.

Through these multiple operational pathways, TraceAlign-DPI addresses all major categories of data plane failures in case of DER outages which is illustrated in detail in Table I. Additionally, if relay and inverter log confirms validity of reported data from DER end after an inconsistency was flagged by TraceAlign-DPI, then it indicates false predictions of TraceAlign-DPI. This observed label is then given as a feedback to the learning module of TraceAlign-DPI to improve its accuracy. While the cyber anomaly tool is responsible for distinguishing attack-induced vs. benign data loss, TraceAlign-DPI and RCA ensure system-level consistency checks and diagnosis by looking into available PS measurements. Rather than considering the flags by TraceAlign-DPI as final anomaly, cyber diagnosis and selective invocation of relay and inverter logs confirm the DER status. This layered approach allows DEROMS to maintain observability and resilience even when data is missing or manipulated and TraceAlign-DPI or cyber anomaly tool gives false positives/ negatives.

3) Processing Methodology: TraceAlign-DPI operates as a continuous integrity monitor by converting incoming measurements into stride-1 overlapping windows of length L ($L = 10$ samples in our testcase). Each window $X_{t-L+1:t} \in R^{L\times F}$ is scored independently by the model, and anomaly scores are aligned to the window end time t. To ensure continuity when data are processed in segments, the last $L-1$ samples of each segment are carried over into the next. The system averages scores from overlapping data chunks to keep them in sync with time. It only triggers an alert if the score stays high for several steps in a row, helping avoid false alarms.

4) DEROMS Waiting Time Parameter t_{wait} for TraceAlign-DPI Invocation: Upon initial detection or suspicion of a DER outage, DEROMS does not immediately trigger the TraceAlign-DPI integrity checker. Instead, a predefined delay interval, denoted as t_{wait}, is introduced to allow the system to stabilize and to account for momentary disturbances that do not constitute actual outages.

The need for t_{wait} is rooted in the dynamic response behavior of inverter-based DERs under abnormal grid conditions, as governed by the IEEE 1547-2018 standard [40]. According to this standard, DERs must support specific grid ride-through capabilities during voltage and frequency excursions, with defined thresholds and response modes.

One critical mode is *cease to energize*, or *momentary cessation*, defined as a temporary suspension of active current exchange between the DER and the area electric power system (EPS), while remaining electrically connected. Besides, IEEE 1547 permits DERs to enter momentary cessation or continue operation under what is termed *permissive operation* mode. The duration of momentary cessation before initiating a trip depends on the DER category:

- *Category I and II DERs:* Up to 0.5 seconds
- *Category III DERs:* Up to 12 seconds

Momentary cessations are not actual outages and typically resolve autonomously. Thus, t_{wait} serves as a buffer to avoid premature RCA or restoration actions in response to temporary ride-through behavior. Also, during genuine DER outages, t_{wait} allows the microgrid to settle into a new steady-state operating point. Premature invocation of TraceAlign-DPI may capture transient effects such as frequency deviations or PQ flow spikes, which can distort its inference regarding the true DER status. To balance responsiveness and accuracy, this work proposes the following values:

- $t_{\text{wait}} = 5$ seconds for Category I and II DERs
- $t_{\text{wait}} = 15$ seconds for Category III DERs

These values are chosen to exceed the maximum permissible momentary cessation windows while providing sufficient time for post-outage transients to settle, ensuring that TraceAlign-DPI operates on stable system data.

5) Discussion on Scalability and Adaptability: With an increasing number of solar DERs, the computational burden of DEROMS may rise. However, the framework is designed as part of the MGMS, which only monitors the DERs under its own territory, primarily focusing on utility-scale solar DERs, since their outages have greater cascading impacts on stability. Thus, assuming a limited number of large DERs per MGMS is both practical and consistent with real deployments. Besides, the TraceAlign-DPI module, though data-intensive, is run periodically rather than continuously, reducing real-time burden. Furthermore, the TraceAlign-DPI tool learns the correlations among system components from their measurements, independent of DER location. This means it inherently captures location-specific effects on system dynamics. The framework is also adaptable to different topologies. Retraining with topology-specific "normal" operating data enables DEROMS to adjust to the unique dynamics of each configuration.

III. Outage Root Cause Analysis Layer

A. Understanding Outage Scenarios Considering Cyber-Physical Failures in Control Plane

To enable accurate and explainable outage diagnosis, the root cause analysis engine must be grounded in a clear taxonomy of DER outage scenarios, including both expected power system behaviors and unexpected cyber-physical failure modes. This subsection outlines the primary causes of DER outages, categorized into normal protection responses and cyber-induced outages where cyber-induced outages may arise from either non-malicious failures (e.g., packet loss, communication dropouts) or adversarial intrusions (e.g., spoofing, replay, or malicious trip commands), both of which are summarized in Table II.

1) Expected Outages Due to Legitimate Protection Mechanisms: These are normal operating responses of DERs to system conditions that exceed safety thresholds as prescribed by IEEE 1547-2018:

TABLE I
DECISION OUTCOMES FOR OUTAGE DETECTION AND CLASSIFICATION INTEGRATING DATA FLAG, DPI RESULT, CYBER CLASS, AND RELAY/ INVERTER LOG

Primary flags & Decisions					Actual Case - Outage						Actual Case - DER active					
Data	**TraceAlign-DPI consistent?**	**Cyber class from packet meta-data?**	**Predicted case**	**Action**	**Comment on Data**	**Comment on DPI**	**Comment on Cyber Anomaly tool**	**Log Find-ings**	**Final class**	**Misclass chance**	**Comment on Data**	**Comment on DPI**	**Comment on Cyber Anomaly tool**	**Log Find-ings**	**Final class**	**Mis-class cha-nce**
Zero/ NaN values - shows possible outage	Yes (outage): system-wide shift/ anomaly detected pointing outage; system AS $> T_p$	OK - supports data & outage	**True outage**	1. Outage → A1	Visible true outage	True positive - correctly detects post outage effect	None	F1, F3, F5	Outage	0%	Falsely reported outage	False positive (un-likely): Falsely detected system-wide shift & linked to outage	(unlikely) - fails to detect critical cyber anomaly	F2, F4, F6, F9	Active	0%
		Anom - questions data & outage	**True outage with cyber anomaly**	1. Outage → A1, 2. A2			(unlikely) mostly non-critical since data reflects system	F1, F3, F5, F7	Outage	0%			usually critical anomaly since data is altered	F2, F4, F6, F8	Active	0%
	No (active): post-outage like system wide shift/ anomaly was not detected, system AS ¡ $> T_p$	OK - supports data & outage	System AS $> T_{ls}$ → **True Outage**, system AS $< T_{ls}$ → **Falsely reported outage**	1. Outage prediction → A1		False negative (un-likely): Fails to detect post outage effect	None	F1, F4, F5	Outage/ active	when post outage system shift too low to be detected (unlikely)		True negative: Does not falsely detect a shift	(unlikely) - fails to detect critical cyber anomaly	F2, F3, F6, F9, F8	Active	0%
		Anom - questions data & outage	system AS $> T_{ls}$ & non-critical cyber anomaly → **True outage**, **Falsely reported outage** otherwise	1. Critical cyber anomaly/ true outage → A1, 2. A2			(unlikely) mostly non-critical since data reflects system	F1, F4, F5, F7	Outage/ active	if post outage system shift too low to be detected or critical cyber anomaly undetected (unlikely)			usually critical anomaly since data is altered	F2, F3, F6, F8	Active	0%
Non Zero values - shows DER is active	Yes (active): post-outage like system wide shift/ anomaly was not detected, system AS ¡ $> T_p$	OK - supports data & active DER	**DER active**	1. A3	Masked outage	False negative (un-likely): fails to detect post outage effect	(unlikely) - fails to detect critical cyber anomaly	not invo-ked	Active	100% since data, DPI, & CA tools all fail	True active status	True negative: Does not falsely detect a shift	None	not in-voked	Active	0%
		Anom - questions data & active DER	**Active DER with cyber anomaly**	1. Critical cyber anomaly → A1, 2. A2			usually critical since data is altered	F1, F4, F6, F8	Outage/ active	If critical cyber anomaly undetected			(unlikely)- mostly non-critical since data reflects system	F2, F3, F5, F7	Active	0%
	No (outage): system-wide shift/ anomaly detected pointing outage; system AS $> T_p$	OK - supports data & active DER	System AS $< T_{us}$ → **DER active**, system AS $> T_{us}$ → **Masked outage**	1. Masked outage prediction → A1		True positive: correctly detects post outage effect (usually strong)	(unlikely) - fails to detect critical cyber anomaly	F1, F3, F6, F9, F8	Outage/ active	when post outage system shift is not very strong		False positive (un-likely): Falsely detected a system wide shift & linked to outage	None	F2, F4, F5	Active	0%
		Anom - questions data & active DER	system AS $< T_{ls}$ & non-critical cyber anomaly → **Active DER**, **Masked outage** otherwise	1. Critical cyber anomaly/ masked outage → A1, 2. A2			usually critical since data is altered	F1, F3, F6, F8	Outage/ active	when post outage system shift is not very strong or critical cyber anomaly undetected (unlikely)			(unlikely)- mostly non-critical since data reflects system	F2, F4, F5, F7	Active	0%

∗ **Thresholds:** T_p = primary anomaly threshold; T_{us} = secondary upper threshold; T_{ls} = secondary lower threshold.
∗ **Actions:** A1 = invoke relay/ inverter log; A2 = Take cyber recovery based on anomaly type; A3 = Invoke DPI in next regular interval.
∗ **Findings from relay/ inverter log:** F1 = Confirms Outage, F2 = confirms active DER, F3 = ground truth strengthens DPI's finding, F4 = ground truth is sent as a feedback to retrain & improve DPI, F5 = Confirms Data reflects system, F6 = Confirms Data does not reflect system, F7 = impact of cyber anomaly as non-critical, F8 = impact of cyber anomaly as critical, F9 = Insight of undetected cyber anomaly.
* Finding F6 replaces DER end data with forecast aware estimated data.

TABLE II
OUTAGE SCENARIOS AND TRACEALIGN-RCA DIAGNOSIS STRATEGIES

Outage Type	**Expected Signatures**	**RCA Strategy / Evidence Alignment**
Voltage/ Frequency Violation	Sudden drop in current, voltage sag or frequency drift; relay trip on threshold violation	PMU transients before outage; relay logs match grid code violation; SCADA confirms breaker open
Overcurrent Fault	Elevated current magnitude prior to trip; breaker open	Relay protection function (overcurrent), PMU shows current spike; match relay trip time with power drop
Islanding	Zero export/import at PCC; frequency or voltage deviation; transfer trip logged	Relay log showing transfer trip; PMU at PCC shows abrupt shift; inverter logs may confirm shutdown
Planned Outage	DER offline; logged remote trip command	Relay/inverter log shows legitimate operator-issued command; SCADA log confirms maintenance schedule
Sensor Failure	DER sees out-of-bound values; no actual system disturbance	Relay triggered by invalid CT/PT values; adjacent PMUs show normal data; SCADA from other DERs unchanged
Breaker Failure	Trip or no-close without protection signal; steady values before trip	PMU/SCADA show normal values before loss; relay log blank or mismatch; breaker fails to follow control logic
Controller/MPPT Failure	Diverging inverter output, unstable MPPT or PLL prior to trip	Inverter logs show internal faults; SCADA shows rapid voltage/current fluctuation before loss; no external cause
Communication Failure	Missed trip/reclose signal; lingering abnormal conditions	Communication heartbeat loss; delay in SCADA command acknowledgment; inverter fails to respond due to setpoint lapse
Cyber Attack (Relay)	DER trips unexpectedly; no real grid event	Relay log shows remote command from unauthorized source; altered settings; log tampering or deletion
Cyber Attack (Inverter)	DER trips due to remote disable or bad setpoints	Inverter logs show unknown command source or force-shutdown; PMU/SCADA show stable pre-outage conditions
Cyber Attack (DoS)	DER fails to reconnect or respond; no new commands received	Logs/metadata show timeout, dropped packets; relay/inverter accessible but unresponsive; SCADA mismatch

- *Voltage and Frequency Violations:* If voltage or frequency deviates outside predefined ride-through thresholds and exceeds the allowable duration, DERs are required to trip. This is common during major grid disturbances or blackouts.
- *Overcurrent Events:* Although inverter-based DERs have limited fault current contributions, faults, particularly ground faults, may induce backfeed currents. Protection relays respond by tripping breakers or commanding inverter shutdown.
- *Islanding Conditions:* In case of upstream grid outage, anti-islanding protection is invoked (e.g., via transfer trip), disconnecting DERs to avoid unintentional island operation.
- *Planned Maintenance Events:* DERs may be deliberately shut down via remote command for maintenance or operational coordination. These events are usually logged and traceable.

2) Unexpected Outages Due to Non-Malicious Cyber-Physical Failures: These failures arise from equipment malfunction, sensor corruption, or communication loss - without active adversary presence:

- *Sensor Failures:* Faulty PTs or CTs may falsely indicate abnormal voltage, current, or frequency - erroneously triggering protection functions. Similarly, falsely reported breaker open status may force the inverter to enter standby or offline mode.
- *Breaker Failures:* Spontaneous tripping or failure to reclose may result in DER disconnection despite acceptable operating conditions.
- *Inverter/Controller Instabilities:* Failures in MPPT tracking, PLL desynchronization, or Q-control divergence can result in internal protection trips.
- *Communication Failures:* DERs relying on external setpoints or reclose/trip commands from the MGMS/DERMS may trip if those signals are lost, delayed, or corrupted due to communication outages.

3) Cyber-Induced Outages From Adversarial Actions: These events are intentionally triggered by attackers aiming to disrupt DER operation through manipulation of control layers:

- *Breaker Compromise:* Attackers remotely issue trip commands or modify protection settings to cause unnecessary DER disconnection.
- *Inverter Hijacking:* By injecting malicious commands or destabilizing setpoints (e.g., PV voltage references, frequency setpoints), attackers can force the inverter into protective shutdown.
- *Denial-of-Service (DoS) Attacks:* Saturating the communication channel or firewall with traffic can block the transmission of essential reclose or operational commands, leaving DERs inoperable.

Each of these scenarios exhibits unique cross-layer signatures that the root cause inference engine must recognize and differentiate. Some are identifiable via consistent phasor and SCADA measurements, while others require deeper analysis of relay logs, inverter shutdown codes, or cyber-metadata anomalies. The accuracy and timeliness of root cause classification depend on the system's ability to detect and reason over these nuanced cyber-physical interactions.

B. Cyber-Aware Outage Root Cause Analysis With TraceAlign-RCA

The TraceAlign-RCA module is designed to diagnose the true cause of a DER outage by aligning heterogeneous evidence across multiple layers of observability - physical measurements

(PMU, SCADA), protection events (relay logs), inverter-side feedback, and cyber-communication metadata. Given that cyber-induced, sensor-induced, and legitimate protection-induced outages can all manifest similarly at the DER output (e.g., power drop), disambiguating these requires domain-specific signal alignment.

Table II summarizes the distinct outage scenarios introduced earlier along with the expected signatures and trace alignment strategies used to infer their cause.TraceAlign-RCA traverses these signal domains, aligns the timing and causality of observed events, and infers the most likely cause of the outage. For example, a fault-induced outage will be corroborated by a spike in PMU current, a relay trip record, and a SCADA breaker status change. In contrast, a sensor-induced false trip will show mismatch between DER-end and adjacent-node data, while cyber-induced outages often leave trails in command metadata, access logs, or abnormal packet structures. By leveraging this cross-domain consistency analysis, TraceAlign-RCA improves outage explainability, enhances operator trust, and enables resilient downstream control decisions.

IV. Outage Response and Recovery Layer

Timely and accurate identification of the root cause of a DER outage is critical not only for system situational awareness but also for enabling a cyber-resilient recovery strategy. Knowledge of the outage origin, whether physical, cyber, or systemic, empowers the MGMS to undertake context-specific responses. For instance, if the outage is confirmed as a hardware fault, appropriate field repair crews can be dispatched with the correct tools, reducing diagnostic delays. If the outage is due to a cyber-attack, containment measures (e.g., isolating compromised relay or inverter units) can be prioritized to prevent lateral propagation. Most importantly, the inferred cause provides a tentative repair time estimate or DER recovery horizon, which is essential for short-term grid reconfiguration, dispatch optimization, and adaptive islanding decisions within the networked microgrid cluster. Without this awareness, the system risks overreacting or underutilizing available flexibility, leading to suboptimal resilience.

A. *Enforcing Scenario Specific Restoration Schemes*

Once the root cause is identified and the DER's operational status is classified (e.g., trusted, unavailable, or compromised), the outage response layer invokes scenario-specific restoration strategies. These are formulated as optimization problems tailored to the physical and cyber-operational constraints of the networked microgrid. Below, we describe several representative scenarios along with their restoration objectives and key variables.

Scenario 1: DER Outage Requiring Repair + Redispatch of Other DERs

In this case, one or more DERs are rendered inoperable due to hardware failure or confirmed cyber compromise. The objective is to minimize the overall operational cost of remaining DERs while also discouraging excess dependency on grid import, especially if the DER was originally deployed as a non-wires alternative (NWA). The modified economic dispatch problem is modeled as:

$$\min_{P_i^{\text{DER}},\, P_{\text{PCC}}} \sum_{i\in\mathcal{G}} C_i(P_i^{\text{DER}}) + \lambda \cdot \left((P_{\text{PCC}})^2 - (P_{\text{PCC,}}\ \text{Rated})^2\right)$$

where: $C_i(P_i^{\text{DER}})$ is the generation cost function of DER i, P_{PCC} is the real power imported from the main grid either at the point of common coupling (PCC) of the MG or at distribution feeder level with λ, being the weighting factor to penalize excess grid import. The optimization constraints include power flow, power balance and security constraints as well as topology constraints if network reconfiguration is permitted with NMGs [48].

Scenario 2: Substation Feeder Outage with DERs Still Operational

Here, the utility substation feeder is unavailable (e.g., grid-side breaker trip), which primarily triggers DER trip due to loss of grid. Since the DERs in MG boundary are healthy, MGMS initiates off-grid mode and through coordination within self or NMG boundary, aims to restore power partially. Restoration focuses on maximizing the supply of critical loads using DER coordination, involving one DER operating in grid-forming mode:

$$\max_{P_j} \sum_{j\in\mathcal{L}_{\text{crit}}} w_j P_j$$

where: P_j is the restored critical load at node j, w_j is the priority weight of load j and $\mathcal{L}_{\text{crit}}$ is the set of critical loads. The MIP problem is subject to power balance equations as well as DER dispatch limits, voltage and current limits, and operational mode constraints for the grid forming agents.

Scenario 3: DER Outage in an Off-grid Microgrid

This case considers an off-grid microgrid that loses a DER asset. Restoration involves potentially collaborating with NMGs for additional support and reallocating available DERs to restore critical loads. DERs deemed operational are included, while unavailable DERs are excluded. Additionally, if the DER facing outage is the grid forming (GFM) asset of the corresponding off- grid Microgrid, then the loss of the DER will result in discontinuation of power supply from other healthy DERs in the MG. The flexibilities of NMG have the potential to utilize these functional DERs in this case by sharing the grid forming capabilities. The restoration strategy must ensure at least one GFM asset is reconnected with all healthy Grid following DERs while maximizing critical loads.

$$\max_{P_j,\mathcal{T}} \sum_{j\in\mathcal{L}_{\text{crit}}} w_j P_j \quad \text{s.t. } \exists\, i \in \mathcal{G}_{\text{GFM}} : i \in \text{Topology}(\mathcal{T})$$

where $\mathcal{G}_{\text{GFM}}$ is the set of grid-forming DERs and Topology$(\mathcal{T})$ denotes active configuration of the microgrid. Constraints ensure that at least one GFM DER is present in the operating island, along with standard load flow and security constraints.

B. *Testbed Development*

V. Implementation and Experimental Setup

The proposed DEROMS framework is validated on a simplified two-microgrid distribution testbed, modeled in RSCAD

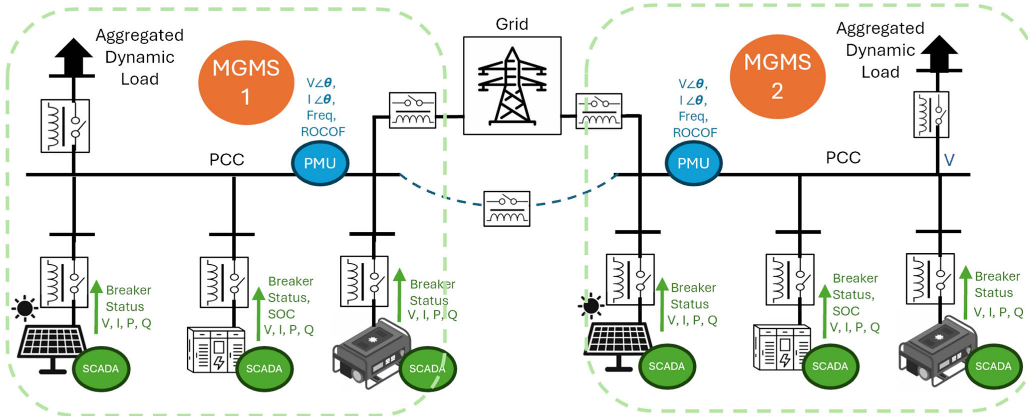

Fig. 2. Layout of the networked microgrid testbed.

and executed in real time using RTDS as shown in Fig. 2. Each microgrid consists of:

- A solar PV system of 400 kW rated generation (V mpp $\sim$ 2 kV, Impp $\sim$ 200 A), integrated with an average inverter model, implementing MPPT and PLL-based grid synchronization, accepting Q setpoints.
- The BESS is modeled with 250 series cells per stack ($\sim$ 975V) and 250 stacks in parallel ($\sim$212.5 Ah), yielding $\sim$ 207kWh total energy capacity, corresponding up to 200kW discharge power for one hour. The BESS is controlled via operator input setpoints P_{bat} and Q_{bat}.
- A grid-forming diesel generator with 1 MW of rated capacity capable of black-start and voltage and frequency reference support.
- The inverters are rated at 1.25 times the DER capacity and invert the DC side voltage to three phase grid voltage.
- Three loads each having 400 kW of peak demand and dynamic load profile, is considered as part of the microgrid. Their cumulative load is modeled with one aggregated dynamic load component (setpoint for $P_{load} \sim 1.2MW$ when all three loads are supplied at their peak demand). The power factor is considered $\sim$ 0.9, making setpoints for Q_{load} around half of P_{load}.
- Relay and breakers are modeled at every DER connection with proper protection mechanisms, which are also capable of receiving remote commands.
- Control system consisting of PI loops for PLL, MPPT, and Q setpoint calibration.

The two microgrids are tied to a single 480 V AC distribution bus, with breakers at each PCC. Each MG connects to the grid through a separate PCC feeder for independent operation and operates in grid-connected mode by default. A tie-line breaker which is normally open allows interconnection to enable networked microgrid operations during islanding or contingency without creating loops.

A. Data Stream and Monitoring Layer

Each DER is monitored through SCADA gateways that provide Analog measurements: Voltage (V), Current (I), Active Power (P), Reactive Power (Q) or SOC levels for battery as well as digital breaker status, streamed at 1 frame/sec. Besides that, PMUs are placed at the PCC of each microgrid which streams Voltage and current phasors ($V\angle\theta$, $I\angle\theta$), frequency, and ROCOF at 60frames/sec via a phasor data concentrator. Following our previous work [49], every 1-second aggregated data window is processed for denoising, outlier removal, and further analysis.

B. Cyber-Physical Attack Simulation

Cyberattack scenarios are implemented using multiple MATLAB-based host agents interfaced with the testbed and a python platform to run the DEROMS. These represent IEDs (Intelligent Electronic Devices) and attacker nodes following the MITRE ATT&CK for ICS framework [50]. Fig. 3 illustrates three representative cyber cases and case 4 is the extreme case of combined data and control plane attack:

- **Case 1: No Cyber Manipulation** – Events occur without any adversarial control; used for baseline comparison.
- **Case 2: Control Plane Attack** – A malicious host (N5) performs a man-in-the-middle (MITM) attack and sends unauthorized trip commands, causing forced DER disconnection (ICS Tactic TA0105: Technique T0831 and T0826).
- **Case 3: Data Plane Attack** – The attacker intercepts and modifies the data stream (e.g., falsifies breaker status, masks actual outages), without affecting physical control (ICS Tactic TA0105: Technique T0832 and T0828).
- **Case 4: Coordinated Control and Data Plane Attack-** A malicious host (N5) executes a sophisticated two-part attack. It first sends an unauthorized trip command to force a DER offline (control plane attack) and simultaneously intercepts and modifies the outbound data stream to falsely report that the DER is still operating normally, thereby masking the forced outage from the operators (data plane attack).

C. Validation of the Proposed DEROMS Framework

To validate the framework, we adopted our previously developed tools as integral detection modules. First, the multivariate LSTM-autoencoder anomaly detection component from our

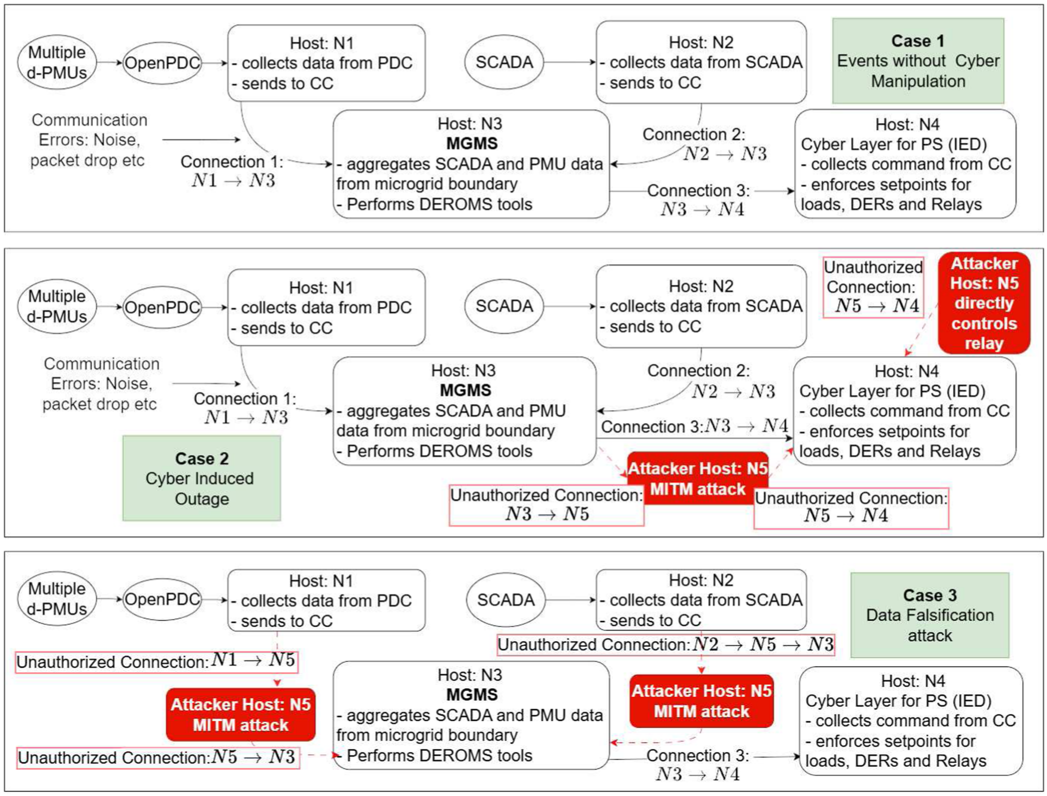

Fig. 3. Cyber-physical scenario simulation encompassing data and control plane.

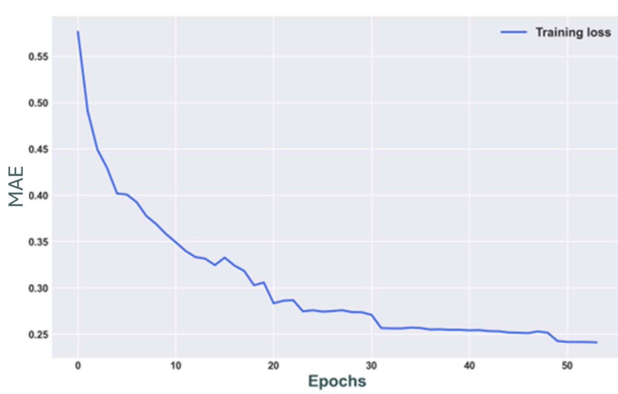

Fig. 4. Training loss history from TraceAlign-DPI.

previous work on "Cyber Anomaly-Aware Distributed Voltage Control with Active Power Curtailment and DERs" framework [51] was embedded into TraceAlign-DPI's architecture. Here, the reconstruction-error scores of the autoencoder on power and voltage deviations trigger cross-domain PQ-flow alignment checks, enabling TraceAlign-DPI to confirm or refute data-layer anomalies against system-wide dynamics [51]. While originally developed for voltage control applications, performance of this tool can be further enhanced through customized design adaptations aligned with the proposed framework, which will be explored in future work. Subsequently, the cyber anomaly detection technique proposed in [52] serves as the cyber anomaly detection tool. Besides, the nowcasting method described in [53], providing a 5 minute forecast, works as the forecasting tool

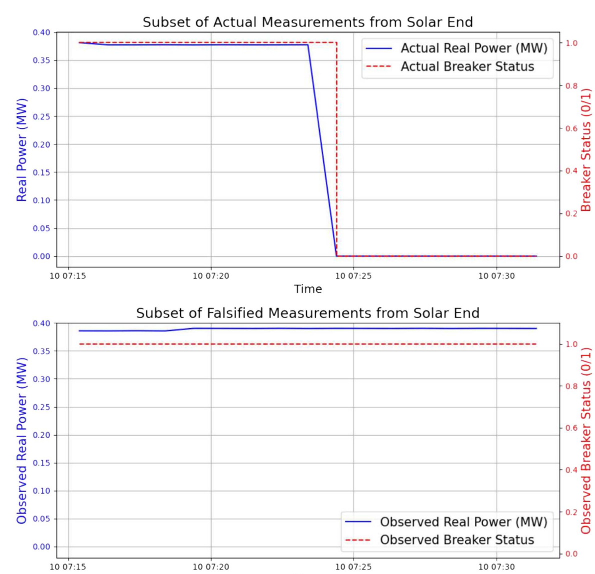

Fig. 5. Actual vs. falsified measurements from solar DER.

and is modified with domain knowledge and a tolerance band of 20 percent to serve as the Forecast-aware estimation tool. Simplified restoration problems are solved using python optimization packages. As part of the microgrid restoration problem, three loads (400 kW peak each, pf $\sim$ 0.9) are treated separately with criticality factors of 1.0, 0.8, and 0.6. The dispatch optimization applies cost and capacity constraints to BESS and DG units, while PV output follows irradiation-driven variability.

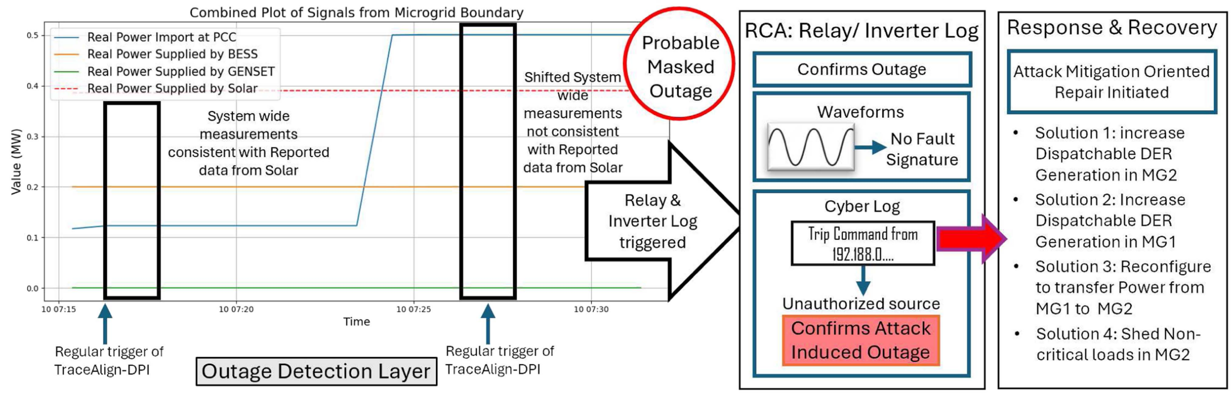

Fig. 6. Workflow of proposed DEROMS for case study 1 (corresponding to cyber-physical attack case 4).

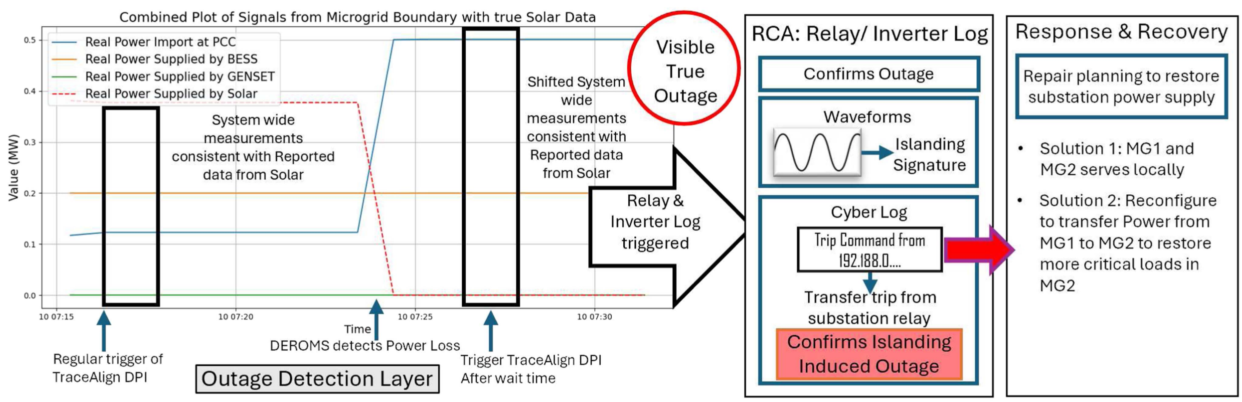

Fig. 7. Workflow of proposed DEROMS for case study 2 (corresponding to cyber-physical attack case 1).

Using the validation dataset, the model achieved Recall of 0.66, Precision of 0.28, and an F1-score of 0.40. The confusion matrix corresponds to a false negative rate (FNR) of 0.34 and a false positive rate (FPR) of 0.60. These results confirm that the model is highly sensitive (successfully detecting two-thirds of anomalies) while leaving room for improvement in the following steps of root cause analysis of the framework. Through invocation of relay/inverter log precision and recall improves to nearly 1. The learning curve shown in Fig. 4 shows the effective training accuracy of the model.

1) Case Study 1: To demonstrate the cyber resilience of the proposed DEROMS framework, Case Study 1 simulates an extreme case of Coordinated Control and Data Plane Attack as referred to cyber-physical attack case 4 (V-B). Here an attacker forcefully disconnects the solar from the system and masks this outage, thus introducing a data plane failure. In our testbed, the solar DER in Microgrid 2 (MG2) undergoes an outage, but the real-time measurements are replaced with historical values as shown in Fig. 5, emulating a replay attack or sensor malfunction. As a result, OMS is not triggered immediately since no power loss is reported from DER. However, the periodic TraceAlign-DPI check detects a post-outage like system shift (e.g., increased import at PCC) and prompts DEROMS to trigger log inspection from the associated relay/inverter. The logs confirm the outage but the waveform analysis reveals no pre-outage fault signature. Instead, the cyber log containing a trip command issued from an unknown IP address, confirms that the outage was attack-induced.

Upon confirmed outage, DEROMS proceeds to the recovery phase. As the root cause is cyber-related, the repair plan focuses on securing the DER's IED and relay access. In parallel, economic dispatch is solved under various cost and dispatch limit constraints resulting in the following solution approaches:

- *Solution 1:* Penalizing substation power supply causes the MG2 diesel generator and BESS to increase output locally.
- *Solution 2:* When MG2 DER dispatch limits are enforced, MG1 compensates by increasing generation and eventually reducing substation power supply.
- *Solution 3:* Penalizing import at MG2 PCC specifically leads to optimized radial reconfiguration from MG1 to share power to MG2.
- *Solution 4:* For tighter operational limits, DEROMS selectively curtails non-critical loads within MG2.

This case demonstrates DEROMS's ability to detect masked outages, confirm cyber-induced trip causes, and apply flexible restoration strategies using the networked microgrid topology as shown in Fig. 6.

2) Case Study 2: In Case Study 2, the DEROMS framework is evaluated under a visible outage scenario where the solar trips due to a grid-side event. The DER power loss is directly observed in the reported measurements, triggering TraceAlign-DPI. The DPI module verifies the outage by matching with system-wide dynamics. This case study corresponds to cyber-physical attack case 1- No cyber manipulation (V-B).

Relay and inverter logs are then queried to determine the root cause. The relay waveform reveals an islanding signature, and the cyber log confirms receipt of a transfer trip command issued by the substation relay, likely due to upstream grid loss. Thus, the outage is classified as an islanding-induced DER trip.

In the recovery layer, the primary repair objective is to restore the grid-side connection. Meanwhile, MGMS-1 and MGMS-2 independently reconfigure and supply local loads using internal DERs. A secondary recovery solution is identified when MG2's load criticality is increased and networked operation is enabled, allowing MG1 to export power and support MG2 via the tie-line, demonstrating adaptive coordination between networked microgrids. The workflow is demonstrated in Fig. 7.

In our implementation, TraceAlign detection runs in the millisecond range, with restoration optimization also completed in milliseconds. Excluding the t_{wait} parameter, the framework generates a decision within 1 s. Perhaps, fast detection is essential to prevent local outages from cascading, and to limit adversarial dwell time during cyber events. While communication and actuation may add latency in practice, the framework remains suitable for real-time microgrid operation.

VI. Conclusions and Future Work

This work presented a novel cyber-resilient DER Outage Management System Framework (DEROMS) designed for solar-rich networked microgrids. The primary contribution of this paper is the theoretical formulation and architectural design of the DEROMS, which includes a multi-layer architecture comprising: (i) real-time outage detection through TraceAlign-DPI, (ii) root cause analysis via TraceAlign-RCA using synchronized physical and cyber traces, and (iii) scenario-aware restoration optimization leveraging the flexibility of networked microgrids. As a critical proof-of-concept, a real-time hardware-in-the-loop testbed was developed to validate the system under both physical faults and cyber-attack scenarios, including data falsification and unauthorized control plane manipulation. Results demonstrated DEROMS's capability to detect visible and hidden outages, infer cyber-physical root causes, and adaptively reconfigure microgrid operations for resilient restoration. By integrating real-time monitoring, data authenticity validation, and adaptive restoration strategies, the system aims to enhance the resilience and reliability of modern distribution systems.

The work presented herein is positioned as a theoretical and methodological foundation upon which more extensive, large-scale experimental analysis must be built. This work mainly focuses on solar DERs, but the DEROMS framework is broadly applicable to other DER types such as wind turbines, BESS, EV chargers, and hybrid inverters. Since the outage root causes, fault signatures, and forecasting techniques are DER-specific, tailoring the analytics to diverse resources remains an important direction for future research. Additionally, periodic invocation of the TraceAlign-DPI tool reduces the real-time computational burden but improvements can be made for faster response, better accuracy and greater robustness against advanced cyber threats. Integrating forecasting techniques, state estimation, and distributed optimization methods will further enhance the system's autonomy and scalability.

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